

Veterinary Package Insert

Product Name:

DEXHEXINE 2% W/W WATER SOLUBLE POWDER

Product Description:

White, crystalline powder which contains 20mg of bromhexine hydrochloride in 1g.

Pharmacodynamics:

Bromhexine is a mucoregulator. By activating the secretion of the seromucous glands, bromhexine helps to re-establish the viscosity and elasticity of bronchial secretions in the tracheobronchial tree. In addition, its expectorant action encourages mobilisation of mucus and enables effective bronchial drainage, thereby improving the functioning and defence capability of the lung. These two simultaneous actions lead to an abundant discharge and facilitate a productive cough.

It breaks down the network of acid glycoprotein fibres found in mucoid sputum, which are mainly responsible for the characteristic viscosity.

Bromhexine modifies the distribution of antibiotics in the organism and increases their concentration in the serum and in the nasal secretions (e.g. spiramycin, tylosin and oxytetracycline). When administered concomitantly with the product, antimicrobial agents should, nevertheless, not be underdosed..

Pharmacokinetics:

Absorption

In Swine, bromhexine is rapidly absorbed following oral administration with a peak plasma concentrations obtained in one to three hours. The concentration plateau is reached 12 hours after the second or third administration.

In cattle, plasma concentrations increase progressively over several hours following administration.

In turkeys or broilers, peak plasma concentrations are reached within 2 to 4 hours of oral administration.

Distribution Due to the lipophilic character of bromhexine, it has a strong affinity for lipid tissues and a slow depletion profile from these tissues.

Metabolism Bromhexine is largely metabolised into more polar compounds.

Elimination The apparent total radioactivity plasma half-life elimination after the last administration is 20 to 30 hours in a pig, 40 to 50 hours in cattle and 40 to 50 hours in chickens and turkeys.

Indications:

Mucolytic treatment of congested respiratory tract.

Recommended Dosage:

To be administered orally

0.45 mg of bromhexine per kg bodyweight daily, equivalent to 2.5g of product per 100 kg bodyweight per day administered for 3 to 10 consecutive days, in drinking water.

In order to obtain the correct dosage the concentration of bromhexine has to be adjusted accordingly.

The following formula may be used to calculate the required concentration of product (in milligrams of product per litre drinking water):

$$\begin{array}{rclcl} 25 \text{ mg of} & & & & \\ \text{the product} & & \text{Average} & & \\ \text{per kg body} & \times & \text{body weight} & = & \dots \text{ mg of the} \\ \text{weight and} & & \text{(kg)} & & \text{product per litre} \\ \text{per day} & & \text{of} & & \text{of} \\ & & \text{animals to be} & & \text{drinking} \\ & & \text{treated} & & \text{water} \\ \hline \text{Average daily water intake} & & & & \\ \text{(l/animal)} & & & & \end{array}$$

The required amount of product should be weighed as accurately as possible using suitably calibrated weighing equipment.

Recommendation for dilution:

- Prepare a relevant quantity of water in a container.
- Add the product to the water while stirring the solution.
- Prepare the solution with fresh water immediately before use.

When using a water proportioner, adjust the pump between 1% to 5% and adapt the volume of preparation accordingly. Do not set up the dosing pump below 1%.

When using a water tank, it is recommended to prepare a stock solution of at least 0.5g of product/L and to dilute it to the target final concentration.

Turn off the water supply to the tank until all the medicated solution is consumed.

For example :

Prepare a stock solution at 0.5 g of product/L, dilute 1/3 to obtain a medicated solution at 0.17g of powder/L eq to 0.5g of powder/3L.

For swine, when administered during the meal, first dissolve the product in water and then add feed. Administration during the meal should be restricted to individual treatment or for treatment of a small group of animals. The preparation must be used immediately. Care should be taken that the intended dose will be completely ingested.

The intake of medicated water depends on the clinical condition of the animals.

Any unused medicated water should be discarded after 24 hours.

Mode of Administration:

Oral

Contraindications:

Do not use in cases of pulmonary oedema.

In case of serious lungworm infection, the drug should only be used 3 days after the commencement of the anthelmintic treatment.

Do not use in cases of hypersensitivity to the active substance or to any of the excipients.

Warning and Precautions:

People with known hypersensitivity to bromhexine or any of the excipients should avoid contact with the veterinary medicinal product.

Avoid contact with skin and eyes.

In case of accidental eye contact, flush the affected eye with copious amounts of clean running water.

Wash hands and exposed skin after administering the product.

Interactions with Other Medicaments

The product may be used in conjunction with antibiotics and/or sulphonamides and bronchodilators.

Bromhexine modifies the distribution of antibiotics in the organism and increases their concentration in the respiratory system.

Pregnancy and Lactation:

Studies in laboratory animals have not produced evidence of foetotoxic effects or effects on fertility at the recommended dose. However this has not been specifically studied in the target species. Use only according to the benefit-risk assessment by the responsible veterinarian.

Adverse Effects:

None known.

Symptoms and Treatment of Overdose:

None known.

Withdrawal Period:

Cattle (calves) - Meat and offal: 2 days

Not permitted for use in cows producing milk for human consumption.

Swine - Meat and offal: Zero days.

Chickens, turkeys and ducks - Meat and offal: Zero days

Do not use in birds producing eggs for consumption, during and 4 weeks before the laying phase.

Packaging:

Packed in 500gm aluminium foil

Storage Condition:

Store below 30°C in dry place, protected from light.

Shelf Life

Two years from date of manufacturing

To be used within 24 hours after first opening of container.

To be used within 24 hours after dilution.

Product Registration Holder

DEXCHEM INDUSTRIES SDN. BHD.

(Company No.: 555717-U)

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Manufactured and Distributed by:

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